Homework 6 — CSCI 347 Data Mining

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> Show your work. Include any code snippets that you used to generate answers. Complete this assignment individually. Unless otherwise stated, you can use code snippets to generate answers. If you are asked to do the calculations by hand, include your working.

In [1]:
```python
import numpy as np
import pandas as pd
from fractions import Fraction
```

Question 1 — Naïve Bayes (Categorical) [15 points]

Suppose you are asked to use Naïve Bayes to predict whether a person will go hiking or not, based on the following **training** dataset.

In [2]:
```python
data = pd.DataFrame({
    'Weather':    ['Snow','Overcast','Sunny','Overcast','Overcast','Snow','Overcast
                   'Snow','Overcast','Overcast','Sunny','Snow','Snow','Windy','Wind
    'Weekend':    ['Yes','No','Yes','Yes','No','No','Yes','Yes','No','No',
                   'Yes','Yes','No','No','Yes','No','No','Yes','Yes'],
    'Finished_HW':['No','No','No','Yes','Yes','Yes','No','No','Yes','Yes',
                   'No','No','Yes','Yes','Yes','No','No','No','Yes'],
    'Go_Hiking':  ['Yes','No','Yes','Yes','Yes','No','No','No','Yes','Yes',
                   'No','No','Yes','Yes','No','No','No','No','Yes']
})
data
```

Out[2]:

	Weather	Weekend	Finished_HW	Go_Hiking
0	Snow	Yes	No	Yes
1	Overcast	No	No	No
2	Sunny	Yes	No	Yes
3	Overcast	Yes	Yes	Yes
4	Overcast	No	Yes	Yes
5	Snow	No	Yes	No
6	Overcast	Yes	No	No
7	Sunny	Yes	No	No
8	Sunny	No	Yes	Yes
9	Snow	No	Yes	Yes
10	Snow	Yes	No	No
11	Overcast	Yes	No	No
12	Overcast	No	Yes	Yes
13	Sunny	No	Yes	Yes
14	Snow	Yes	Yes	No
15	Snow	No	No	No
16	Windy	No	No	No
17	Windy	Yes	No	No
18	Windy	Yes	Yes	Yes

Part (a) — Prior Probabilities [4 points]

Find **P(c_i)**, the prior probabilities for the "Yes" and "No" classes, where c_1 = "Yes" and c_2 = "No".

P(c1=Yes) = 9/19 = 0.47

P(c2=No) = 10/19 = 0.53

Part (b) — Likelihoods $P(x|c_1)$ and $P(x|c_2)$ [8 points]

Given the new data instance **x = (Sunny, Yes, No)**, find $P(x|c_1)$ and $P(x|c_2)$.

In [7]:
```
# Your work here
x = {'Weather': 'Sunny', 'Weekend': 'Yes', 'Finished_HW': 'No'}
print("New Instance:", x)
```

```
New Instance: {'Weather': 'Sunny', 'Weekend': 'Yes', 'Finished_HW': 'No'}
```

P(x|c1) = (3/9)(4/9)(2/9) = 0.033

P(x|c2) = (1/10)(6/10)(8/10) = 0.048

Part (c) — Posterior Scores [2 points]

Compute **$P(x|c_1)\cdot P(c_1)$** and **$P(x|c_2)\cdot P(c_2)$**.

P(x|c1)·P(c1) = 0.033 · 0.47 = 0.0156

P(x|c2)·P(c2) = 0.048 · 0.53= 0.0253

Part (d) — Classification [1 point]

Which class should the new data instance be assigned to?

Since 0.0253 > 0.0156, x is classified as "No"/will not go hiking

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Question 2 — Gaussian Naïve Bayes [4 points]

Use Gaussian Naïve Bayes for the following numerical training dataset. Predict the class of the new data instance **x**.

In [4]:
```
numerical_data = pd.DataFrame({
    'Rain_inches': [0, 0.5, 1, 5, 0.3, 0.4, 0.1, 0, 0, 3, 6, 2.1, 1.02],
    'Sleep_hours': [9, 5, 7, 7, 8, 4, 9, 9, 8, 10, 8, 8, 8.5],
    'HW_pct':      [80, 90, 95, 100, 100, 100, 27, 50, 100, 98, 95, 70, 98],
    'Go_Hiking':   ['Yes','No','Yes','Yes','Yes','No','No','No','Yes','Yes','No','N
})
numerical_data
```

Out[4]:

	Rain_inches	Sleep_hours	HW_pct	Go_Hiking
0	0.00	9.0	80	Yes
1	0.50	5.0	90	No
2	1.00	7.0	95	Yes
3	5.00	7.0	100	Yes
4	0.30	8.0	100	Yes
5	0.40	4.0	100	No
6	0.10	9.0	27	No
7	0.00	9.0	50	No
8	0.00	8.0	100	Yes
9	3.00	10.0	98	Yes
10	6.00	8.0	95	No
11	2.10	8.0	70	No
12	1.02	8.5	98	Yes

In [8]:

```python
# New data instance
x_new = (0.58, 5.3, 75)
features = ['Rain_inches', 'Sleep_hours', 'HW_pct']

def gaussian_pdf(x, mean, std):
    return (1 / (std * np.sqrt(2 * np.pi))) * np.exp(-0.5 * ((x - mean) / std) ** 2

yes = numerical_data[numerical_data['Go_Hiking'] == 'Yes']
no  = numerical_data[numerical_data['Go_Hiking'] == 'No']
n   = len(numerical_data)

score_yes = len(yes) / n
score_no  = len(no)  / n

for i, feat in enumerate(features):
    score_yes *= gaussian_pdf(x_new[i], yes[feat].mean(), yes[feat].std())
    score_no  *= gaussian_pdf(x_new[i], no[feat].mean(),  no[feat].std())

print(f"p(x|Yes) * P(Yes) = {score_yes:.10f}")
print(f"p(x|No)  * P(No)  = {score_no:.10f}")
```

```
p(x|Yes) * P(Yes) = 0.0000008210
p(x|No)  * P(No)  = 0.0001282006
```

X is classified as No/will not go hiking. This is because P(x|Yes)·P(Yes) < P(x|No)·P(No)

Question 3 — Association Rule Mining [11 points]

Consider the following data that shows transactions of items purchased in a supermarket.

In [6]:
```
transactions = {
    1:  ['Toilet paper', 'Beans', 'Rice', 'Milk', 'Baby wipes', 'Diapers'],
    2:  ['Oat milk', 'Beans', 'Toilet paper', 'Orange juice', 'Bread'],
    3:  ['Oat milk', 'Milk', 'Orange juice', 'Toilet paper', 'Bread'],
    4:  ['Beans', 'Toilet paper', 'Baby wipes', 'Diapers'],
    5:  ['Toilet paper', 'Butter', 'Baby wipes', 'Diapers'],
    6:  ['Milk', 'Toilet paper', 'Bread'],
    7:  ['Milk', 'Rice', 'Bread'],
    8:  ['Beans', 'Milk', 'Rice', 'Toilet paper'],
    9:  ['Milk', 'Butter', 'Diapers', 'Bread'],
    10: ['Beans', 'Rice', 'Toilet paper'],
}

for tid, items in transactions.items():
    print(f"T{tid:2d}: {items}")
```

```
T 1: ['Toilet paper', 'Beans', 'Rice', 'Milk', 'Baby wipes', 'Diapers']
T 2: ['Oat milk', 'Beans', 'Toilet paper', 'Orange juice', 'Bread']
T 3: ['Oat milk', 'Milk', 'Orange juice', 'Toilet paper', 'Bread']
T 4: ['Beans', 'Toilet paper', 'Baby wipes', 'Diapers']
T 5: ['Toilet paper', 'Butter', 'Baby wipes', 'Diapers']
T 6: ['Milk', 'Toilet paper', 'Bread']
T 7: ['Milk', 'Rice', 'Bread']
T 8: ['Beans', 'Milk', 'Rice', 'Toilet paper']
T 9: ['Milk', 'Butter', 'Diapers', 'Bread']
T10: ['Beans', 'Rice', 'Toilet paper']
```

Part (a) — Support of {Milk, Toilet paper} [2 points]

What is the support of the itemset **{Milk, Toilet paper}**?

4/10 = 0.4

Part (b) — Frequent Itemsets of Size 2 (minsup = 4) [3 points]

If we use the minimum support threshold of **4**, what are the itemsets of size 2 that are frequent?

{Beans, Toilet Paper}(5 occurences)

{Milk, Bread}(4 occurences)

{Milk, Toilet Paper}(4 occurences)

Part (c) — A-Priori Candidate Itemsets of Size 3 (minsup = 4) [4 points]

If we use the **A-Priori algorithm** to generate frequent itemsets using a minsup of 4, what is the set of **candidate itemsets of size 3** that will be generated by the algorithm?

{Beans, Toilet Paper} + {Milk, Toilet Paper} -> {Beans, Milk, Toilet Paper} is pruned because {Beans, Milk} is not frequent.

{Milk, Bread} + {Milk, Toilet Paper} -> {Milk, Bread, Toilet Papper} is pruned because {Bread, Toilet Paper} is not frequent

Therefore C_3 = empty set

Part (d) — Confidence & Lift of rice → beans [2 points]

What is the **confidence** and **lift** of the rule **rice** → **beans**?

Confidence = 3/4 = 0.75

Lift = 0.75/0.5 = 1.5

Part (e) — Confidence & Lift of beans → rice [1 point]

What is the **confidence** and **lift** of the rule **beans** → **rice**?

Confidence = 3/5 = 0.6

Lift = 0.6/0.4 = 1.5